

LOCAL ORGANIZATION OF BIOLOGICAL MEMBRANES MODULATES TAU-LIPID INTERACTIONS AND FIBRIL FORMATION

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ABSTRACT

Tau is a protein found in the brain where it regulates microtubule activity. It is an intrinsically disordered protein, which means that it does not have a unique three-dimensional structure. Tau can form highly ordered aggregates that are directly involved in several neurodegenerative diseases, called tauopathies, including Alzheimer's disease. Recent research has shown that Tau protein can interact directly with the neuronal membrane, mainly composed of lipids. Observations of Alzheimer's brain tissues show the association of Tau fibers with membrane lipids such as phosphatidylcholine, cholesterol, and sphingolipid.

Yet, how membranes modulate tau aggregation pathways remains poorly understood. We present a biochemical and biophysical characterization of tau amyloid filament formed in presence of different membrane models. We used Electron Paramagnetic Resonance (EPR) to directly quantify the population of tau bound to the membrane and to quantify the affinity between Tau and lipids. After showing the importance of the electrostatic interaction between Tau and anionic lipids, we investigate how the amount and density of charges influence Tau aggregation. This work allows us to draw a general model where membrane-induced Tau aggregation is a two-step process. First, the binding to the membrane through electrostatic interactions is a necessary but not a sufficient step. Second, the nucleation of Tau amyloids at the membrane surface occurs only when specific conditions are fulfilled, namely, high surface density altering Tau conformation and spatial proximity between aggregation-prone conformers. A direct implication of this model is that local membrane heterogeneities, such as phase separation or lipid rafting, are strong modulators of Tau aggregation. This work provides the molecular basis to predict how different membrane states regulate Tau aggregation.